

Code report

By Team: Go With The Flow

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This program is a discrete-time simulation of TCP's sliding window protocol, demonstrating both congestion control (TCP Tahoe) and flow control. A sender node and receiver node exchange frames and acknowledgments across a simulated lossy channel, where frames and ACKs may be corrupted with a configurable probability. The sender adjusts its congestion window (cwnd) based on incoming ACKs and timeout events, while the receiver advertises its available buffer space (rwnd) on every ACK. After the simulation completes, three matplotlib graphs visualize the congestion window over time, the interaction between flow control and buffer usage, and the cumulative ACKs received by the sender.

How to run it

The program requires Python 3 and the matplotlib library. Install matplotlib with:

pip install matplotlib

- 1.) Open a terminal of your choice (Command Prompt, terminal, etc.)
- 2.) Navigate to the directory containing tcp_simulation.py
- 3.) Run the command: `python tcp_simulation.py`
- 4.) Input your parameters... Following the template below:
"Input Parameters: <duration: int> <sws: int> <rws: int> <error: int> <timeout: int>"

5.) Press Enter Example Test: `C:\directory_to_python_file>python tcp_simulation.py`
Input parameters: 16 6 6 0.05 3
Parameters: duration=16 sws=6 rws=6 error=0.05 timeout=3

After the simulation runs, the program prints a tabular log of every timestep to the console and displays a three panel matplotlib graph.

Output

The simulation was run with the following parameters: duration=300, sws=16, rws=4, error=0.05, timeout=8. The full output spans 300 timestamps. The excerpt below shows timestamps 25-50 that are printed from the terminal.

Each row of the output represents a single timestep. The columns are organized as:
timestamp | frame_sent | lar | lfs | lfr | laf | ack | buffer

```
| 24  ?  24  | 23  27  ?  ()
```

26		25	23	25		24	28	24	()
27		26	24	26		25	29	25	()
28		27	25	27		26	30	26	()
29		28	?	28		27	31	?	()
30		29	?	29		28	32	?	()
31		30	28	30		29	33	29	()
32		31	29	?		?	33	—	()
33		32	29	32		29	33	29	(31)
34		—	29	32		29	33	29	(31, 32)
35		—	29	32		29	33	—	(31, 32)
36		—	29	32		29	33	—	(31, 32)
37		—	29	32		29	33	—	(31, 32)
38		—	29	32		29	33	—	(31, 32)
39		30	29	32		29	33	—	(31, 32)
40		31	29	32		32	36	32	()
41		33	32	33		32	36	32	()
42		34	32	34		33	37	33	()
43		35	33	35		34	38	34	()
44		36	34	36		35	39	35	()
45		37	35	37		36	40	36	()
46		38	36	38		37	41	37	()
47		39	37	39		38	42	38	()
48		40	38	40		39	43	39	()
49		41	39	41		40	44	40	()
50		42	40	42		41	45	41	()

Next that is printed is the graphs.

